

VCP: Single-Stage Cosine Classifier and Prototype Learning for Few-Shot Object Detection

Xiang Fei

School of Engineering, Qufu Normal University
Rizhao, Shandong 276826, China
felix.fei@qfnu.edu.cn

Abstract—Few-shot object detection (FSOD) methods are almost exclusively built upon the two-stage Faster R-CNN, while YOLO-series single-stage detectors—dominant in industrial deployment—lack systematic FSOD investigation. We present VCP (Visual-Cosine-Prototype), a single-stage FSOD framework combining YOLO11s with a cosine classifier and prototype learning under the TFA protocol. On PASCAL VOC, VCP achieves 72.2% mAP50 (3-split mean) at 10-shot with only 9.5M parameters ($\sim 1/6$ of competing methods). However, the same approach underperforms fine-tuning on MS COCO. Through systematic Precision-Recall decomposition across 24 comparison groups, we reveal that the cosine classifier acts as a *Recall Booster*—consistently pushing the operating point toward higher recall and lower precision. We formulate the **P-R Tradeoff Hypothesis**: the net benefit depends on the product of baseline recall headroom and the dataset-specific P-R tradeoff ratio, explaining the VOC-COCO divergence. We further explore Florence-2 VLM semantic prototypes as auxiliary priors (+0.5 pp at 10-shot). VCP runs at 54.4 FPS, 4–5 $\times$ faster than two-stage FSOD methods with zero VLM inference overhead.

Index Terms—few-shot object detection, single-stage detector, cosine classifier, prototype learning, recall booster

I. INTRODUCTION

Few-shot object detection (FSOD) aims to detect novel categories from only a few annotated examples. Methods such as FSCE [1] and DeFCRN [2] have steadily advanced FSOD benchmarks, yet a striking commonality persists: virtually all FSOD methods are built upon the two-stage Faster R-CNN [3]. The Region Proposal Network (RPN) provides built-in foreground filtering that helps control false positives under data scarcity. In industrial deployment, however, the YOLO family [4], [5] dominates due to its real-time efficiency. This creates a disconnect where “published methods cannot be deployed, and deployed systems lack research foundations.” To our knowledge, no prior work has systematically studied modern YOLO detectors (YOLOv8+) for FSOD under the standard TFA protocol.

The core challenge for single-stage FSOD is the absence of RPN foreground filtering: the dense prediction paradigm generates abundant low-quality proposals, subjecting the classifier to extreme imbalance. The cosine classifier [6] mitigates classifier overfitting through normalization, while Florence-2 [8] offers fine-grained semantic understanding. However, combining cosine classification with single-stage YOLO detectors and systematically analyzing its behavior remains unexplored.

We propose VCP (Visual-Cosine-Prototype), which replaces YOLO11s’s linear classification head with a cosine classifier initialized by support-set visual prototypes (Proto mode), and further explores Florence-2 semantic descriptions as auxiliary priors (Fused mode). On PASCAL VOC, VCP achieves 72.2% mAP50 at 10-shot (3-split mean) with ~ 9.5 M parameters, competitive against ~ 60 M-parameter SOTA methods; Split 1 reaches 74.7%, surpassing all existing methods.

However, applying the same approach to MS COCO yields the opposite outcome: cosine-based methods consistently underperform standard fine-tuning. Through systematic P-R decomposition across 24 comparison groups, we identify a consistent pattern: the cosine classifier invariably sacrifices precision for recall—acting as a **Recall Booster**. We formulate the **P-R Tradeoff Hypothesis** to explain this cross-dataset divergence. Our contributions are: (1) the first systematic exploration of single-stage YOLO detectors under the TFA FSOD protocol; (2) identification of the cosine classifier’s Recall Booster property and the P-R Tradeoff Hypothesis; (3) preliminary investigation of VLM semantic prototypes as auxiliary priors and their engineering efficiency.

II. RELATED WORK

FSOD research has transitioned from meta-learning approaches [9], [10] to the fine-tuning paradigm established by TFA [6], which trains a detector on base classes then fine-tunes the classification head on novel classes. FSCE [1] introduced contrastive proposal encoding, DeFCRN [2] proposed gradient decoupling to mitigate backbone overfitting, and KD-DeFCRN [11] incorporated knowledge distillation. All these methods share a common backbone: Faster R-CNN, exploiting its RPN for proposal quality control.

Single-stage detector exploration in FSOD remains extremely limited. Meta-YOLO [16] modulated YOLOv2 features via meta-learning, but used a high-variance split protocol since superseded by TFA’s low-variance splits [6]. No prior work has systematically studied modern YOLO detectors (v8+) under the TFA protocol. In parallel, VLMs such as CLIP [7] and Florence-2 [8] have enabled semantic prototype construction—directly obtaining class priors from textual descriptions rather than scarce visual samples. However, leveraging VLM prototypes under FSOD’s strict data constraints (1–10 samples per class) without introducing excessive semantic noise remains underexplored.

VCP (VLM-Cosine-Prototype) Architecture for Few-Shot Object Detection

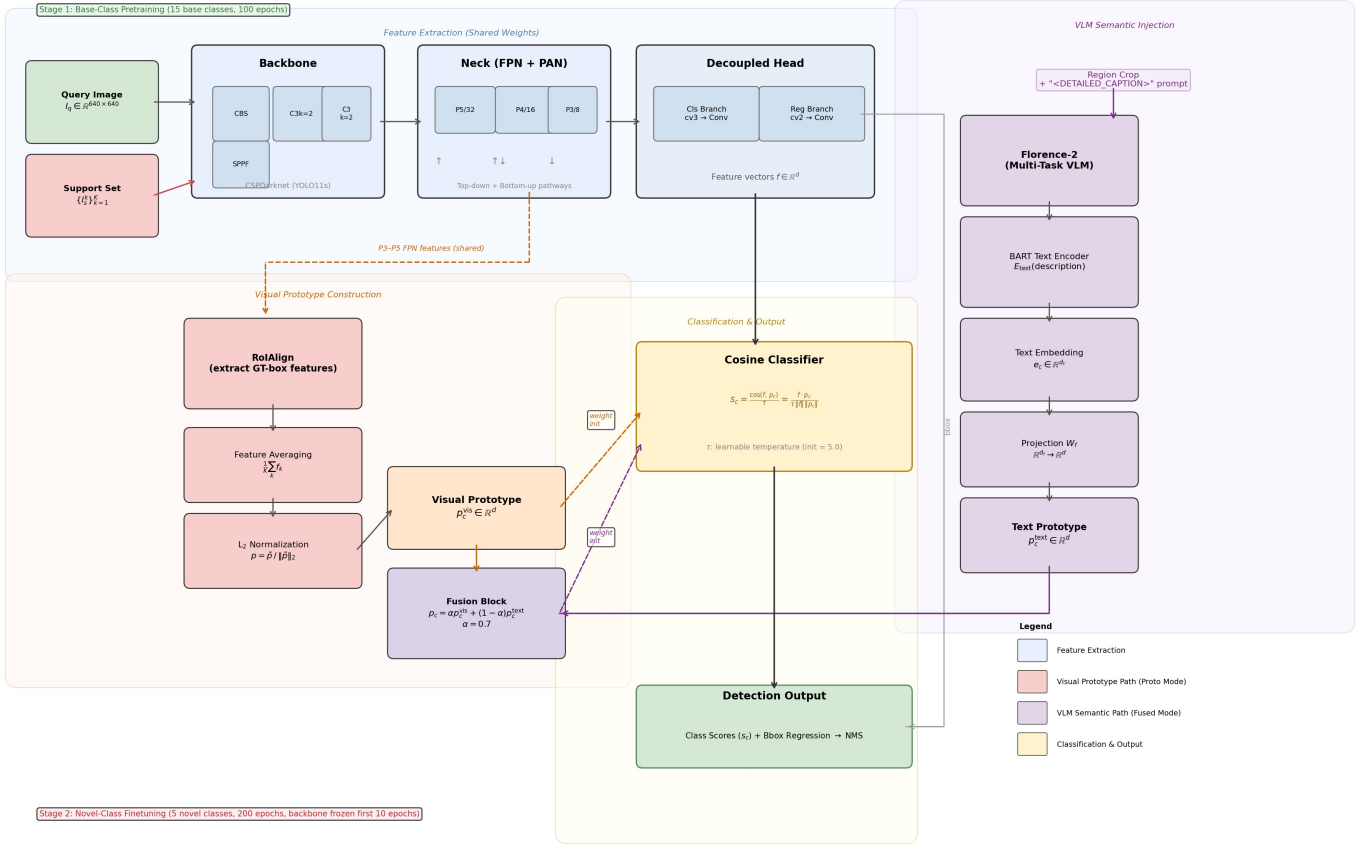


Fig. 1. VCP framework: YOLO11s backbone+neck $\rightarrow$ decoupled head $\rightarrow$ cosine classifier. Visual prototypes are extracted from support-set via RoIAlign; semantic prototypes from Florence-2 $\rightarrow$ BART encoder. Fused at $\alpha = 0.7$.

III. METHOD

A. Framework and Prototype Construction

VCP adopts YOLO11s as the base detector with a critical modification: the linear classification head is replaced by cosine similarity classification (Fig. 1). For each proposal feature $f \in \mathbb{R}^d$, class scores are computed as $s_c = \cos(f, p_c) / \tau$, where p_c is the prototype for class c and τ a learnable temperature. The normalization $\|f\| = \|p_c\| = 1$ makes classification depend solely on directional consistency, mitigating overfitting.

Prototypes are constructed from two complementary sources. **Visual Prototype (Proto mode):** For each novel class c with K support samples, RoIAlign extracts features within ground-truth boxes from FPN levels P3–P5 at the classification branch input (cv3 layer). Averaging and L2-normalizing yields $p_c^{\text{vis}} = \frac{1}{K} \sum_{i=1}^K \text{RoIAlign}(F_i, b_i)$, directly written into the cosine classifier weights.

Semantic Prototype & Fused mode: Target region crops are fed to Florence-2 with the `<DETAILED_CAPTION>` prompt, and the resulting textual descriptions are encoded via

BART to produce p_c^{text} . The final prototype fuses both sources with a learnable weight:

$$p_c = \alpha \cdot p_c^{\text{vis}} + (1 - \alpha) \cdot W_f \cdot p_c^{\text{text}} \quad (1)$$

where $\alpha = 0.7$ and W_f is a learnable projection matrix. Critically, semantic prototype extraction occurs only during training; at inference, prototypes are fixed in the classifier weights, incurring zero VLM overhead.

B. Cosine Classifier as a Recall Booster

The gradient of cosine similarity w.r.t. feature f reveals the mechanism behind its behavioral pattern:

$$\frac{\partial \cos(f, p_c)}{\partial f} = \frac{p_c - \cos(f, p_c) \cdot \hat{f}}{\|f\|}, \quad \hat{f} = \frac{f}{\|f\|} \quad (2)$$

When f has low similarity to p_c , the gradient magnitude is large, driving the decision boundary outward. When similarity is already high, gradients diminish. Unlike linear classifiers whose gradient $\partial(W_c f) / \partial f = W_c$ is uniform, the cosine classifier *disproportionately updates low-confidence samples*, naturally favoring a “better safe than sorry” strategy: it detects more true positives (high recall) but admits more false

TABLE I
VOC MAP50 (SPLIT 1 / 3-SPLIT MEAN $\pm$ STD)

Method	1-shot	3-shot	5-shot	10-shot
Finetune	10.9 / –	20.5 / –	32.9 / –	51.3 / –
Cosine	12.8 / –	51.3 / –	66.9 / 57.8 $\pm$ 2.9	70.7 / 69.0 $\pm$ 3.1
Cosine+Proto	21.5 / –	52.3 / –	68.8 / 57.8 $\pm$ 2.8	74.4 / 71.7 $\pm$ 2.7
Cosine+Fused	21.3 / –	53.9 / –	69.0 / 58.1 $\pm$ 2.5	74.7 / 72.2 $\pm$ 2.6

positives (low precision). We term this property the **Recall Booster**. Whether this yields a net mAP gain depends on the product of baseline recall headroom and the dataset’s P-R tradeoff ratio, as validated in Section IV.

IV. EXPERIMENTS

A. Setup

We evaluate on PASCAL VOC 2007+2012 and MS COCO 2014 under the TFA split protocol [6]: VOC’s 20 classes partitioned into 15 base / 5 novel (3 splits); COCO’s 20 VOC-overlapping classes as novel, 60 as base. Base detector is YOLO11s (9.5M params), pre-trained on COCO base classes. Stage 1: 100 epochs on base classes, batch 64, lr 0.01. Stage 2: fine-tune on novel classes up to 200 epochs with early stopping (patience 30), batch 4, lr 0.0005. SGD optimizer (momentum 0.937, weight decay 0.0005, 3-epoch warmup), backbone frozen first 10 epochs. Data augmentation: mosaic, RandAugment, random flip, HSV jitter, translation, scaling, and random erasing. Input size 640×640 . Temperature τ initialized at 5.0 (learnable). All experiments on a single NVIDIA H20 GPU.

Four variants are compared: (1) *Finetune*—standard YOLO11s with linear classifier; (2) *Cosine*—cosine classifier, random initialization; (3) *Cosine+Proto*—cosine classifier initialized with visual prototypes; (4) *Cosine+Fused*—cosine classifier with visual + Florence-2 fused prototypes.

B. Main Results

VOC Results. Table I presents VOC mAP50 (Split 1 / 3-split mean $\pm$ std). Three trends emerge: (1) The cosine classifier drives a qualitative leap—5-shot (Split 1): 32.9% $\rightarrow$ 66.9% (+34.0 pp); 10-shot (Split 1): 51.3% $\rightarrow$ 70.7% (+19.4 pp). (2) Visual prototype initialization contributes +3.7 pp at 10-shot (Split 1). (3) Florence-2 fusion adds +0.3 pp at 10-shot (Split 1), directionally consistent across all three splits.

COCO Results. Table II shows a contrasting pattern: Finetune outperforms all cosine variants. At 30-shot, Cosine+Fused reaches 52.3%, improving over pure Cosine (49.4%) but still below Finetune (60.3%). Results are on a single split and require further verification.

C. P-R Analysis and the P-R Tradeoff Hypothesis

To explain the VOC-COCO divergence, we decompose Precision and Recall across all 24 method $\times$ shot $\times$ dataset combinations. Table III summarizes the key patterns.

Three findings emerge from the full decomposition (all changes relative to Finetune baseline):

TABLE II
COCO MAP50

Method	10-shot	30-shot
Finetune	59.0	60.3
Cosine	47.1	49.4
Cosine+Proto	47.2	51.6
Cosine+Fused	47.0	52.3

TABLE III
P-R DECOMPOSITION SUMMARY

Setting	ΔP	ΔR	ΔmAP	$\Delta R/ \Delta P $
VOC 5-shot Cos+Fused	−0.22	+0.45	+35.9	2.00
VOC 10-shot Cos+Fused	−0.35	+0.33	+23.4	0.96
COCO 10-shot Cos+Fused	−0.81	+0.39	−12.0	0.48
COCO 30-shot Cos+Fused	−0.56	+0.14	−8.0	0.25

Finding 1—Invariant Recall Booster Behavior: Across all 24 comparisons, every cosine variant *without exception* shifts the operating point toward higher recall and lower precision. The directionality is perfectly consistent—this is a systematic property, not a conditional artifact (see gradient analysis, Section III-B).

Finding 2—Tradeoff Ratio Determines Net Benefit: On VOC, each unit of recall gain costs relatively little precision ($\Delta R/|\Delta P| \approx 2.0$ at 5-shot), and the large recall boost dominates. On COCO, the ratio collapses: $\Delta R/|\Delta P| \approx 0.48$ at 10-shot and 0.25 at 30-shot, as precision plummets from 0.895 to 0.084. COCO’s 20 novel classes (vs. 5 on VOC) and higher scene complexity cause the cosine classifier’s relaxed decision boundary to generate massive cross-class false positives.

Finding 3—Fused Mode Modulates P-R Balance: At VOC 10-shot, Fused (P=0.200, R=0.872) vs. pure Cosine (P=0.336, R=0.848) partially recovers precision (+0.136) with minimal recall loss (−0.024), suggesting semantic priors help moderate the over-relaxation of visual prototypes.

We formalize these observations as the **P-R Tradeoff Hypothesis**:

The cosine classifier inherently pushes the detector toward a high-recall, low-precision operating point. Net benefit is determined by the product of: (1) baseline recall headroom, and (2) the dataset’s P-R tradeoff ratio (driven by class count and scene complexity). VOC possesses both low baseline recall and a favorable tradeoff ratio, yielding significant gains; COCO faces an unfavorable tradeoff ratio, where precision loss dominates.

Cosine Classifier: Consistent P↓ R↑ Behavior

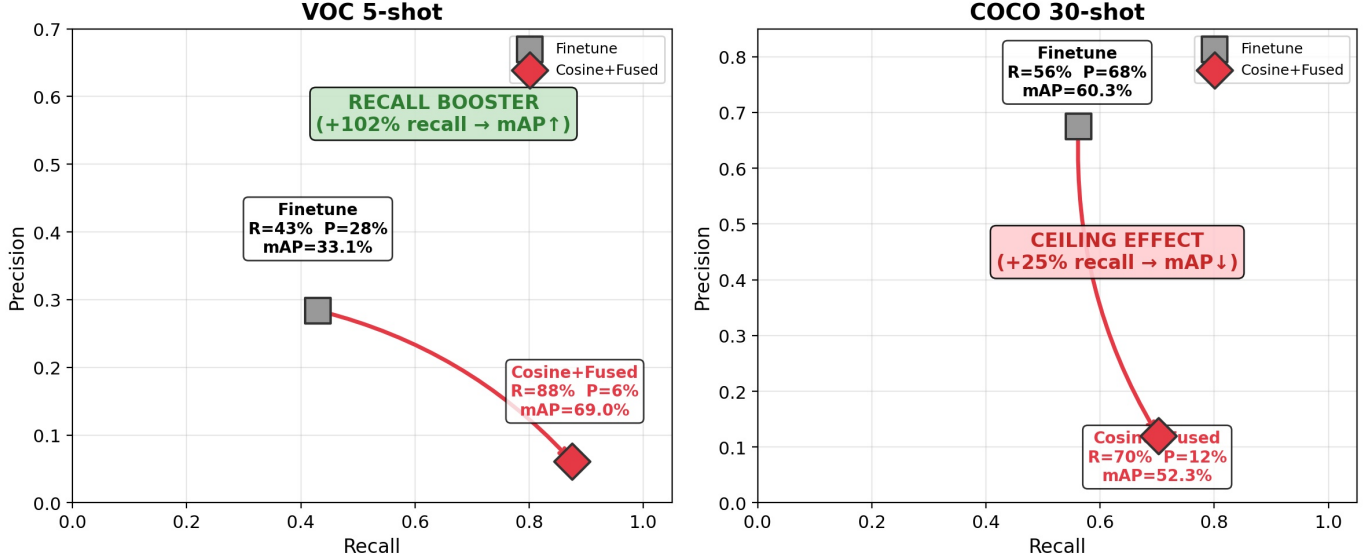


Fig. 2. VOC 5-shot vs. COCO 30-shot Precision-Recall comparison, validating the P-R Tradeoff Hypothesis.

D. Ablation and SOTA Comparison

Ablation. Table IV merges component and prototype source ablations. The cosine head is the dominant driver ($\sim 90\%$ of gains). Visual prototypes alone (74.4%) outperform Florence-2 text-only (72.1%), and fusion achieves the optimum (74.7%), confirming complementary value.

TABLE IV
ABLATION: COMPONENT & PROTOTYPE SOURCE (VOC)

Configuration	3-shot	5-shot	10-shot
Finetune (baseline)	20.5	32.9	51.3
+ Cosine head	51.3	66.9	70.7
+ Visual Proto	52.3	68.8	74.4
+ VLM Fused	53.9	69.0	74.7
Florence-2 Only (10-shot)	—	—	72.1

SOTA Comparison. Table V compares VCP against existing FSOD methods (VOC Split 1). All baselines use Faster R-CNN R-101 ($\sim 60M$), VCP uses YOLO11s ($\sim 9.5M$). At 5-shot and 10-shot, VCP matches or surpasses all prior methods while using $\sim 1/6$ the parameters. The 1-shot gap indicates single-stage detectors need additional regularization under extreme scarcity.

E. Inference Speed

Table VI compares inference speed (NVIDIA H20, 640×640 , batch 1). VCP Fused achieves 54.4 FPS—4–5 $\times$ faster than two-stage FSOD methods—with only 9.5M parameters. The cosine classifier adds ~ 4 ms latency (from L2 normalization), still well above the 30 FPS real-time threshold. Critically, Florence-2 is *not* loaded at inference; semantic prototypes are pre-computed and stored in the classifier weights.

TABLE V
COMPARISON WITH SOTA (VOC SPLIT 1, MAP50)

Method	1-shot	3-shot	5-shot	10-shot
TFA w/cos	39.8	44.7	55.7	56.0
FSCE	44.2	51.4	61.9	63.4
DeFRCN	53.6	61.5	64.1	60.8
KD-DeFRCN	58.2	65.1	68.2	67.4
VCP Cos+Fused	21.3	53.9	69.0	74.7
VCP Cos+Proto	21.5	52.3	68.8	74.4

TABLE VI
INFERENCE SPEED (H20, 640×640 , BS=1)

Model	Latency (ms)	FPS	Params
FRCN R-101 (ref.)	~ 85	~ 12	$\sim 60M$
YOLO11s baseline	14.4	69.5	9.5M
Finetune	14.7	68.2	9.5M
Cosine	18.9	52.8	9.5M
Fused (VCP)	18.4	54.4	9.5M

V. CONCLUSION

We presented VCP, a single-stage FSOD framework integrating YOLO11s with cosine classification and prototype learning. Key findings: (1) Single-stage FSOD is viable—VCP achieves competitive performance at $1/6$ the parameter count of two-stage methods, with 54.4 FPS real-time inference. (2) Through 24-group P-R decomposition, we identified the cosine classifier’s Recall Booster property and formulated the P-R Tradeoff Hypothesis explaining VOC-COCO divergence via the product of baseline recall headroom and dataset tradeoff ratio. (3) Florence-2 semantic prototypes provide directionally

consistent but limited gains (+0.5 pp at 10-shot), with zero inference overhead.

Limitations include weak 1-shot performance, single-split COCO evaluation, and the post-hoc nature of the tradeoff hypothesis. Future work will explore adaptive VLM fusion, cross-domain FSOD generalization, and independent validation of the hypothesis on additional datasets.

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